

Information loss of fixed health indicators as a control problem: simplex geometry and reachability limits

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Abstract

A health indicator compresses several condition-monitoring channels into one scalar, and normally keeps the same weights for the life of the component. Because degradation runs through several mechanisms at differing rates, the channel combination carrying most information about remaining life rotates as the component ages, so a fixed indicator necessarily discards information. We show that the operating point sets how much it discards, which makes that loss a control variable. The loss reduces to a single angle: the angle between the indicator and the direction along which information currently arrives. Writing each channel's share of that information as a probability vector places that angle on a statistical simplex. Zero loss is attainable exactly when the directions reachable at each instant share a common point, which for generic plants needs one fewer independently adjustable operating variable than there are mechanisms. Scaling every channel by the same factor leaves the loss unchanged, so no scalar summary of the Fisher information matrix can detect it. On a representative three-mechanism plant, steering the operating point cuts the discarded information by a factor of 68, and one dimensionless ratio, computable before the designer fixes any indicator, indicates whether a gain of that order is plausible. Across six public run-to-failure fleets the informative direction turns 30.7° over the life of a lithium-ion cell fleet and, against a negative control, 439 simulated turbofan records confirm that the operating point moves it. The same fleets bound the theory: the highest-fidelity turbofan data reject the separable plant form under which the operator's authority becomes computable offline, so that branch describes a model class rather than measured machinery.

Keywords: prognostics, health indicator, remaining useful life, information loss, optimal control, reachability

1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record;

the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster; it also shifts the balance between the mechanisms, and therefore moves the direction in which information about remaining life arrives.

The operating point is, in most applications, something the operator chooses. So the misalignment between a fixed indicator and the informative direction is not fixed by nature. It is a function of the input, and the operator can steer it.

That reframing turns a data-analysis question into a control question, and control questions come with sharper answers than design heuristics do. One can ask what mis-

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alignments are reachable at a given operating envelope,¹⁰⁰ whether zero loss is attainable at all, how many indepen-¹⁰¹ dent inputs that would take, and what the best achievable loss is when it is not. These are the questions this paper¹⁰² answers.¹⁰³

They sit against five bodies of work. Four address parts¹⁰⁴ of the problem and the fifth supplies the geometry used¹⁰⁵ in Section 3; to our knowledge none of them addresses the¹⁰⁶ position taken here.¹⁰⁷

The prognostics literature treats the compression as a¹⁰⁸ design problem to be solved from data: searching ge-¹⁰⁹ netically over candidate features against monotonicity¹¹⁰ and trendability criteria [2], learning it with a recurrent¹¹¹ network [5], or surveying the resulting pipelines [1, 6].¹¹² Two limits on that programme are known, information-¹¹³ theoretic bounds on what any fixed compression can re-¹¹⁴ tain [3] and the number of run-to-failure units needed to¹¹⁵ learn one reliably [4]. The first is our own, and the bound-¹¹⁶ ary is worth drawing explicitly: it takes one degradation¹¹⁷ trajectory as given and asks how much information a fixed¹¹⁸ compression must lose along it. The question here is the¹¹⁹ complementary one, whether that trajectory can itself be¹²⁰ steered so that the same fixed map loses less. Both prior¹²¹ limits hold the trajectory fixed; neither asks whether it¹²² can be changed.¹²³

The predictors downstream are overwhelmingly stochas-¹²⁴ tic — Wiener, gamma and inverse-Gaussian degrada-¹²⁵ tion processes with first-passage failure [7, 8], reviewed¹²⁶ for remaining-life estimation in [9, 10] and embedded in¹²⁷ system-level design methodology in [11]. That literature¹²⁸ models how damage evolves, not how the measurement¹²⁹ should be compressed before the model is applied, which¹³⁰ is the question upstream of it.¹³¹

A separate literature does treat the operating point as a¹³² decision variable, but to extend remaining life rather than¹³³ to preserve estimability [12, 13]; the two aims can conflict,¹³⁴ and Section 6 exhibits an instance in which they disagree.¹³⁵ Choosing inputs to make a state easier to estimate is itself¹³⁶ classical, going back to dual control [14], and its modern¹³⁷ forms shape a scalarisation of the Fisher information ma-¹³⁸ trix [15, 16, 17]. Our Proposition 4.6 shows the quantity¹³⁹ at issue here to be invisible to every such scalarisation.¹⁴⁰

The fifth body is information geometry. The reformu-¹⁴¹ lation of Section 3 rests on the Fisher–Rao geometry of¹⁴² the probability simplex and on the Bhattacharyya coef-¹⁴³ ficient [18, 19], and the flow it produces is the replica-¹⁴⁴ tor equation of evolutionary dynamics [20, 21]. These are¹⁴⁵ standard objects; what is new here is their identification,¹⁴⁶ with the informative direction of a degrading component,¹⁴⁷ and the fact that the fitness driving the flow is actuated.¹⁴⁸

The contributions are as follows.

1. The information a fixed indicator discards is an ex-¹⁴⁷ act function of one misalignment angle (Lemma 2.1).¹⁴⁸ Placing the channels’ shares of the Fisher information¹⁴⁹ on a statistical simplex turns the degradation into a¹⁵⁰ replicator flow whose rates the operating point mod-¹⁵¹

ulates, and the optimal indicator into a barycentre of the degradation path (Section 3).

2. Zero loss is attainable exactly when the reachable di-
rection sets share a point, which generically needs
at least as many independent operating variables as
mechanisms less one (Section 4). Supplying the miss-
ing input collapses the peak reachability floor from
4.52° to zero at every state, so the rule prices in-
strumentation rather than only bounding it. Taking
logarithms makes every reachable set a translate of
one fixed set, which turns the question into a contain-
ment and shows the counting rule to be sufficient but
not necessary; the exceptions exist for any number
of mechanisms and are too fragile to build on (Sec-
tion 4.1). The count separates two objectives that are
easily conflated: holding the misalignment at a con-
stant level is a single scalar condition and generically
needs one input, whereas freezing the informative di-
rection outright is $d - 1$ conditions. That gap is the
difference between useful steering and lossless steer-
ing, and an installation with one controllable variable
can have the first without the second.
3. The loss is invariant to a common gain on all chan-
nels and is therefore invisible to every scalarisation of
the Fisher information matrix, so observability-driven
criteria cannot be used to choose an indicator.
4. The best achievable loss is the value of an optimal
control problem whose objective is a ratio, and whose
terminal time is itself controlled (Section 5), and a
dimensionless regime index, computable before any
indicator is designed, screens for the regime in which
steering can have substantial value (Section 6.7).
5. The premises the construction rests on are separately
testable on public run-to-failure data, and Section 7
tests them in three stages: whether the informative
direction rotates at all, whether the operating point
moves it, and whether the separable stress model sur-
vives. Two survive and one does not, which makes
that section a test of which hypotheses withstand con-
tact with data rather than a demonstration that the
method works. The rotation is present in measured
fleets; the actuation premise is so far confirmed on
simulation only; and higher-fidelity turbofan data re-
ject the separability of Lemma 4.3, which places that
lemma, and the offline route it opens, as statements
about a model class rather than about measured ma-
chinery.

Section 2 states the model. Sections 3 to 5 develop the theory, Section 6 exercises the three of its claims that are stated sharply enough to be refuted, and Section 7 reports what public fleet data can and cannot establish about the premises the theory rests on.

2. Problem formulation

This section states the degradation model, says precisely what a prognostic algorithm is being asked to estimate and where the uncertainty in that quantity comes from, derives the direction along which information about it arrives, and defines the fixed indicator whose information loss the rest of the paper studies. Readers familiar with [3] will recognise the model; the derivation is given in full here because the geometry of Section 3 depends on several of its details.

2.1. Degradation driven by a controlled operating point

A component degrades through d distinct mechanisms acting at once. In a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-electrolyte interphase and mechanical fatigue of the electrode; in a rolling bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the damage accumulated by mechanism i up to time t , and collect these into the state $x = (x_1, \dots, x_d)$.

One externally chosen operating point drives all d mechanisms. It collects the temperature, the mechanical or electrical load, and the duty cycle at which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

with $\mathcal{U}$ compact, representing the operating envelope the application permits. Throughout this paper u is the control input: it is the decision variable, and the question we ask is what choosing it well can buy. The number of independent inputs, m , turns out to decide what is achievable, and it is usually much smaller than the number of mechanisms d .

The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

in which the two factors play different roles:

- $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much faster that mechanism runs at operating point u than at a reference point. It carries all the dependence on the input.
- $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends on the damage it has already done. It carries all the dependence on the state.

This separation is the standard accelerated-life form and covers the stress models in routine use [22]. In the Arrhenius model the input is absolute temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's constant. In the inverse-power-law model the input is a mechanical or electrical stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress exponent. Choosing h_i recovers the exponential, power-law and stretched-exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and $h_i > 0$.

The within-unit dynamics are deterministic, where most of the degradation literature places the randomness inside a single unit's path [7, 8, 10]. We put it between units instead (Section 2.2). The object of study is the direction along which information arrives, which is fixed by the relative rates of the competing mechanisms: a within-unit diffusion would jitter that direction without altering its systematic motion, whereas dispersion in the rate parameters across a population is what makes remaining life uncertain and therefore worth estimating.

We measure damage on a scale normalised so that mechanism i exhausts the component's tolerance when x_i reaches unity. A component is withdrawn at a condemnation limit at or below that point, so we write the end of life as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

No result below depends on the value of x_f , which enters only by fixing where a trajectory stops. The computed quantities of Section 6 do depend on it, and the value used there is stated with the rest of the test plant.

2.2. What is to be estimated, and why it is uncertain

The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

the time left before failure.

We must say where the uncertainty in R lies, because (2) is a deterministic differential equation: for one component whose initial damage and parameters were known exactly, T_f and hence R would be known exactly, and there would be nothing to estimate. The uncertainty is between components, not within one. Units built to the same specification differ in initial damage and in the coefficients of (2), which is the unit-to-unit variability that reliability engineering treats as the primary source of scatter in life data [22]. A prognostic algorithm observes one unit drawn from that population and must infer its remaining life.

We therefore treat R , at any given observation instant, as a random variable over the population of units, with prior variance $\text{Var}[R] > 0$. This variance is the uncertainty the algorithm exists to reduce, and it is the reason the estimation problem is not trivial. Everything below is written for a nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

2.3. Observations and the informative direction

Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal variables. The record available to the

algorithm is the collection of these measurements over the observation window.

Following [3], we linearise the dependence of the record on R about the nominal trajectory, which makes the record and R jointly Gaussian. The sensitivity of channel i to remaining life follows from (4): because R falls at unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

Dividing each channel by its own noise level puts them on a common scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = [\sigma_i^{-1} \varphi_i(u) h_i(x_i)]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (7)$$

In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the *informative direction* and the combination a health indicator has to be aligned with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed indicator depends only on the second. Experiment design in the observability tradition scores the first, which is why Proposition 4.6 finds it blind to the loss.

We use two properties of (7) repeatedly below.

Positivity. Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i > 0$ in (2). This is a consequence of the degradation model rather than an assumption imposed for convenience, and Remark 3.1 shows that the results below would fail without it.

Actuation. $\mathbf{d}$ is an explicit algebraic function of the state and the input. The direction along which information about R arrives is therefore not a fixed property of the instrumentation: it moves when the operating point moves. That is the observation this paper is built on, and Fig. 1 places it: the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

Because R is scalar, a single signal-to-noise ratio governs the information the record carries about it, namely the squared sensitivity times the prior uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

Rates in (7) are taken with respect to clock time, whereas [3] differentiates with respect to age normalised by T_f . The two conventions differ by a factor common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters the objective, restores the normalised convention.

2.4. The fixed health indicator and its information loss

A health indicator compresses the d measured channels into one scalar before any life prediction is made. We take it to be a fixed linear map. Since (7) already measures each channel in units of its own noise, we write the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and $\tilde{y} := \Sigma^{-1/2} y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole life of the component. Fixing it is not a modelling convenience: the map is typically embedded in firmware or frozen by certification, so it cannot be re-tuned as the component ages. An indicator specified on the raw measurements, $z = \mathbf{w}^\top y$, is the same map under $\mathbf{v} \propto \Sigma^{1/2} \mathbf{w}$, and the normalisation $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters because the angle below is measured after whitening: unless the σ_i are equal, a unit vector in raw coordinates is not a unit vector in whitened ones, and the two readings of ψ disagree.

We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. After whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude $\|\mathbf{d}\| \sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise, so the indicator's signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

against the γ of (8) available to an oracle that is free to re-aim its projection at $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio s carries $\frac{1}{2} \log(1+s)$ nats about R [23], the information the fixed indicator forfeits, at each instant, is

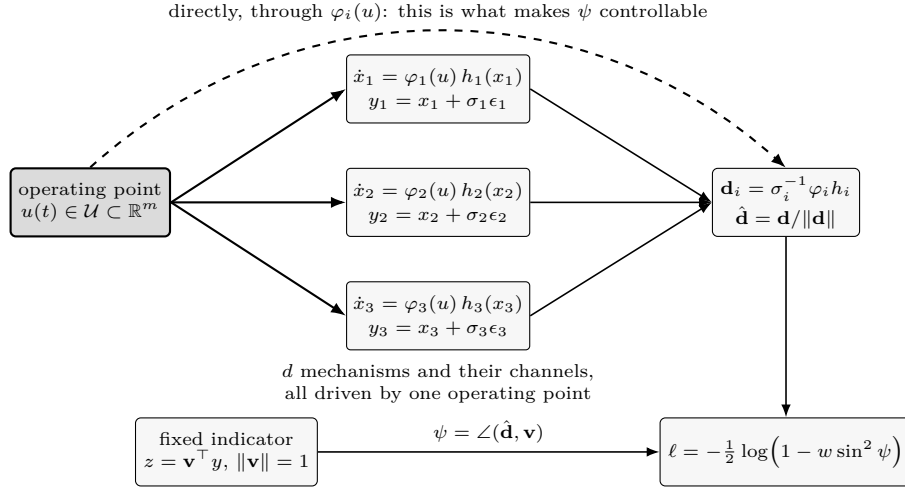
$$\ell = \frac{1}{2} [\log(1+\gamma) - \log(1+\gamma \cos^2 \psi)] \geq 0, \quad (11)$$

The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

a loss per unit time. We optimise this rate rather than the information the whole observation record forfeits, because (12) is unaffected by how densely the record is sampled and is therefore a property of the indicator rather than of the measurement schedule. The two are different quantities, and Proposition 2.2 gives the second in closed form.

Two remarks on what (11) does and does not say. It is a loss relative to an oracle that no fixed map can realise, so it measures the price of the time-invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the single angle ψ : two indicators that make



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

the same angle with $\hat{\mathbf{d}}$ are equally good, whatever else distinguishes them.

The next lemma replaces (11) by a closed form that will be more convenient throughout, and records how it behaves at small misalignment.

Lemma 2.1. *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

exactly and at every signal-to-noise ratio, and

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

Proof. Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log(1 - w \sin^2 \psi).$$

For (14), use $-\frac{1}{2} \log(1 - s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 - \psi^4/3 + O(\psi^6)$; collecting ψ^4 terms gives $-w/6 + w^2/4 = w(3w - 2)/12$. $\square$

The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is therefore the leading term of an exact identity rather than an approximation of unknown quality. The quartic coefficient $w(3w - 2)/12$ changes sign at $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic model overstates the loss, above it the model understates it. This matters in practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life measurement with a weak signal on one side, a well-instrumented component near failure on the other. Fig. 2(a) plots the exact loss against its quadratic approximation on both sides of that ratio.

Proposition 2.2. *Let the record consist of measurements (5) at instants $t_1, \dots, t_N$, with the noise independent across channels and across instants, and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1 + G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

Then the information about R that the fixed indicator forfeits over the whole record is

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (15)$$

Proof. In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an oracle reading the whole record obtains $\frac{1}{2} \log(1 + G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k) R + \mathbf{v}^\top \epsilon_k$, whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| = 1$, so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 + G(1 - S)) = -\frac{1}{2} \log \frac{1 + G - GS}{1 + G} = -\frac{1}{2} \log(1 - WS)$. $\square$

Equation (15) is (13) again, with the instantaneous signal-to-noise ratio γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore

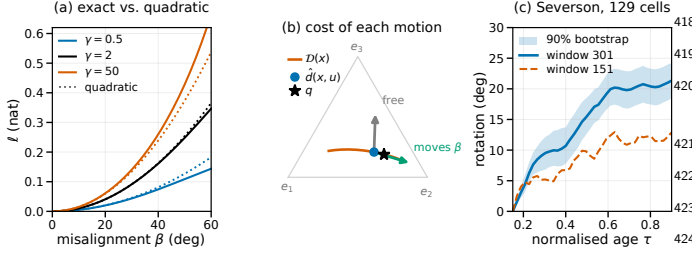


Figure 2: (a) Lemma 2.1: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

survives the passage from an instant to a record. What does not survive is the averaging: $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ. They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top (\sum_k \mathbf{d}_k \mathbf{d}_k^\top) \mathbf{v}$, so the best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$ and should not be read as a statement about (15).

3. The information simplex and its replicator flow

Lemma 2.1 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what follows — the replicator form of Theorem 3.4 and the misalignment rate of Theorem 3.6, which Section 4 builds the reachability argument on. The Bhattacharyya identity and the barycentre are interpretation: they say what the loss and the optimal indicator already are in recognised terms, and nothing later depends on them.

3.1. Information shares

Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

The interpretation is direct. The total Fisher information the record carries about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i is exactly the share of the total information about remaining life that arrives through channel i . A component in which one mechanism dominates the degradation has p near a vertex of the simplex; one in which three mechanisms contribute equally sits near its centre.

The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an optimal indicator may be taken with non-negative weights.

Remark 3.1 (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$ at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$ inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a third of the time. The results therefore require the positive-orthant structure of the degradation model; it is not only a notational convenience.*

3.2. The loss as a Bhattacharyya distance

The identification below is used twice in what follows: it gives “the average of a curve on the simplex” an unambiguous meaning (Section 3.3), and it lets the high-signal limit of the loss be read off as a known divergence.

Proposition 3.2. *$\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya divergence,*

$$\ell = -\frac{1}{2} \log[(1 - w) + w e^{-2D_{\text{B}}(p, q)}]. \quad (17)$$

As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

Proof. By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$ is the classical isometry between the simplex under the Fisher–Rao metric and the positive orthant of the unit sphere, under which the geodesic distance is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2} d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$, and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

The square root is a bijection between the simplex and the non-negative part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition it is an isometry. We use the two descriptions interchangeably from here on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable direction, and a distance between points of the simplex is the angle between the corresponding unit vectors.

The resulting picture is the one the rest of the paper develops. The degrading component is a curve on the simplex, the health indicator is a single point on it, and the information lost depends only on the distance between them. Design of the indicator is the choice of one point; control of the operating point is the shaping of the curve.

3.3. The optimal indicator as a barycentre

If the indicator is one point and the plant is a curve, the best point should be some kind of average of the curve. It is, but of two different kinds in two different limits.

Proposition 3.3. *Let q^* minimise the time-averaged loss.*

- (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ pointwise, so q^* tends to the Bhattacharyya barycentre $\arg \min_q \int D_B(p_\tau, q) d\tau$.*
- (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} := \min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

Proof. (i) As $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (13) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_B$$

by Proposition 3.2. The convergence is uniform, which is what carries the minimisers with it. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$ gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$

is bounded away from zero uniformly in q and in τ and D_B stays finite. Both objectives are continuous on a compact set and uniformly close on it, so their minimisers converge.

(ii) By (14), $\bar{\ell} = \frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu = w d\tau/\bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the definition of the weighted Fréchet mean, and $\psi = \frac{1}{2}d_{\text{FR}}$ converts $\frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2]$ into $\frac{1}{8}\bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$. $\square$

The two limits are taken in different parameters, signal-to-noise ratio in (i) and arc length in (ii), and they do not coincide. On a curve of angular span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5} degrees of the latter by 50 dB. For design purposes the practical consequence is mild: either barycentre of a nominal degradation path is a good starting indicator.

3.4. The replicator flow and the role of the input

We now ask how fast the informative direction moves, and what the operating point does to that motion.

Theorem 3.4. *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i . Along (2),*

$$\dot{\mathbf{d}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

so the information shares obey the replicator equation of evolutionary dynamics, in which each share grows in proportion to the amount by which its own rate exceeds the average, the average being taken over the shares themselves.

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

and

$$\|\dot{\mathbf{d}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = [\varphi_i(u)h'_i(x_i)]_i + \nabla_u \log \varphi \dot{u}. \quad (20)$$

Proof. From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \dot{\mathbf{d}}^\top \dot{\mathbf{d}})$ has components $\dot{\mathbf{d}}_i \dot{g}_i - \dot{\mathbf{d}}_i \sum_j \dot{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives $\text{Var}_p[\dot{g}]$. Since $p_i = \dot{\mathbf{d}}_i^2$, $\dot{p}_i = 2\dot{\mathbf{d}}_i \dot{\mathbf{d}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i \dot{h}_i$, and $(h'_i/h_i)\varphi_i \dot{h}_i = \varphi_i \dot{h}'_i$. $\square$

Equation (20) has a compact reading. The speed at which the informative direction rotates equals the information-weighted spread of the mechanisms' logarithmic rates. If all mechanisms accelerate at the same proportional rate the direction does not move at all, however fast the component is degrading; it is disagreement among the mechanisms that rotates it. The identity is the Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's fundamental theorem of natural selection [18, 20, 21]. What is new here is its identification with the informative direction of a degrading component, and, decisively, that the quantity playing the role of fitness, the logarithmic rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's lever.

Remark 3.5 (contrast with Oja's flow). *Oja's flow for on-line principal-component extraction has the same algebraic form; for a diagonal covariance it reduces to a replicator equation with fitness twice the eigenvalues [24]. The roles are opposite. There the flow is an algorithm, run to fixation in order to find a direction, with constant fitness. Here it is the plant, its fitness is time-varying and actuated, and the aim is to prevent fixation from carrying the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\dot{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.1 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\mathbf{d}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\mathbf{d}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\mathbf{d}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}})/\sin \beta$ and $\dot{\mathbf{d}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\mathbf{d}}, \hat{e} \rangle = \dot{\mathbf{d}}^\top \mathbf{v}/\sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\dot{\mathbf{d}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits below hold for every

measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d - 2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4. Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures the actuation authority available at x , and it shrinks to a point when the mechanisms respond identically to the input.

Theorem 4.1. *Fix an admissible input and let $x(\cdot)$ be the trajectory it generates. Zero loss along that trajectory is attainable with the fixed indicator q if and only if $q \in \bigcap_t \mathcal{D}(x_t)$. Solving $\hat{\mathbf{d}}(x, u) = q$ for u imposes $d - 1$ independent scalar equations on m unknowns, so for generic data a common point exists only if $m \geq d - 1$. If $m < d - 1$, then for generic data $\beta_{\min} > 0$ along the trajectory; Remark 4.5 shows the qualifier cannot be dropped. In every case $\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt / T_f$ for that trajectory.*

Proof. By Lemma 2.1, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$ is the image of an m -dimensional set, hence of dimension at most m , and is a null subset of the $(d - 1)$ -dimensional sphere that a generic q misses; if $m \geq d - 1$ the implicit function theorem supplies a solution wherever the Jacobian has full rank. The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by definition of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$, which is generic. Since ℓ is increasing on $[0, \pi/2]$, the bound follows on dividing by T_f .

This is a state-conditioned result. The bound is genuine in the sense that no input violates a reachability constraint

at the state it occupies, but it is conditional on the trajectory, which the input also steers. Optimising over the trajectory as well is deferred to Section 5, and the two levels are the structure of the argument: a state-conditioned limit here, a trajectory-level limit there.

The practical reading of Theorem 4.1 is a counting rule: a component degrading through d mechanisms generically needs $d - 1$ independently adjustable operating variables before a fixed indicator can be kept perfectly aligned. Section 4.1 says what the qualifier covers. Three mechanisms and a single temperature setpoint will not do it. But a weaker and much cheaper objective is available.

Corollary 4.2 (what actuation is actually required). *Two objectives that are easily conflated carry different requirements.*

- (i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$ conditions on $\dot{u}$, hence generically $m \geq d - 1$.*
- (ii) *Holding β at a constant level, not necessarily zero, is the single scalar equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$ stays in $\mathcal{U}$; generically $m \geq 1$ suffices.*

Proof. (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant; by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap between them is the content of that theorem: $\dot{\beta}$ sees only the component of $\dot{\hat{\mathbf{d}}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by (ii) and constrained by (i). $\square$

Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the trajectory that carries it, and is the problem of Section 5.

4.1. The reachable sets are translates of one set

Theorem 4.1 counts equations against unknowns and hedges with “generic”. The hedge is not idle, and the following observation says exactly what it covers.

Lemma 4.3 (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned} \xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \end{aligned} \quad (23)$$

both taken modulo the all-ones vector. Consequently every reachable direction set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.

Proof. Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along the all-ones vector. $\square$

The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator's authority can therefore be computed offline from the accelerated-life model, with no degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage shapes h and the noise levels σ , and every reachable set is that one set carried along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the plant's demand $\mathcal{A}$ sticks out of the operator's authority $\mathcal{B}$.

That offline route is the most useful consequence drawn in this paper and also the most restrictive. It holds on the separable subclass and not beyond it, and the best high-fidelity turbofan data we have reject separability (Section 7). This subsection and Proposition 4.4 should therefore be read as describing a structured subclass that is convenient when it applies, not as a general account of real machinery. Nothing stated before this subsection depends on the assumption – Theorem 4.1 included – and Fig. 6 marks the one branch of the development that does.

Proposition 4.4 (exact criterion). *Zero loss is attainable along a trajectory if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve in the $(d-1)$ -dimensional quotient fails generically unless $m \geq d-1$; this recovers Theorem 4.1. When it does not fail, $m < d-1$ suffices.*

Proof. By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$ for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

The containment is drawn in Fig. 3(c), and the mechanism the count obscures is now visible. Alignment at an instant fixes u_t outright, since $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint on the state must then hold along the flow with no freedom left to enforce it. One input can carry the whole burden only when no residual constraint is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

Remark 4.5 (a family with $m = 1$ and any d). *Take exponential shapes $h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors. Under the ramp $u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 = \varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Integrating the construction for $d = 2$ to 8 with one input holds the misalignment below 1.5×10^{-6} degrees and the mean loss below 10^{-15} . That is against 1.2×10^{-5} to 5.4×10^{-4} for the best constant input on the same plant.*

This is a statement about the sharpness of Theorem 4.1 and not a way to save an actuator. The family of Remark 4.5 is a knife edge. Perturbing the shape coefficients

ν by 1% raises the mean loss to 3.2×10^{-4} under the myopic policy, against 1.8×10^{-4} for the best constant input on the same perturbed plant. At one percent of shape error, steering is already worse than not steering. Closing the loop halves the damage but does not repair it, so the fragility belongs to the construction and not to the open-loop schedule. Mistuning the activation energies instead is tolerated to about two percent. A designer should read Proposition 4.4 as an explanation of when the counting rule binds, and Theorem 4.1 as the rule to design against.

4.2. Invisibility to Fisher-information criteria

A natural objection is that all of this should follow from designing the input to maximise the Fisher information about R , as in the observability-aware literature discussed in Section 1. It does not, and the reason is structural.

Proposition 4.6 (Fisher blindness). *The loss of a fixed indicator is not a function of the Fisher information about R : configurations with equal Fisher information can have different losses, and configurations with equal losses can have different Fisher information. No criterion depending on the Fisher information alone can therefore rank fixed indicators by the loss they incur.*

Proof. For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$, so trace, determinant and least eigenvalue coincide and every scalarisation is a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$, that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain therefore equalises the scores of two configurations with different losses, and separates the scores of two configurations with the same loss. $\square$

Stated in engineering terms, a Fisher criterion measures the strength of the signal, whereas the loss of a fixed indicator depends on its direction. Amplifying every channel by the same factor increases the first and leaves the second unchanged. Section 6 exhibits this concretely: with the Fisher score held equal to seven digits, the loss still spans a factor of 9.4.

5. The best achievable loss as a fractional control problem

Theorem 4.1 bounds the loss along a given trajectory. But the input steers the trajectory as well as the misalignment on it, so a fundamental limit requires optimising over both at once. This changes the structure of the problem.

The objective is a time average over a lifetime that the input itself determines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes (24) a fractional optimal control problem, meaning that the objective is a ratio rather than that the

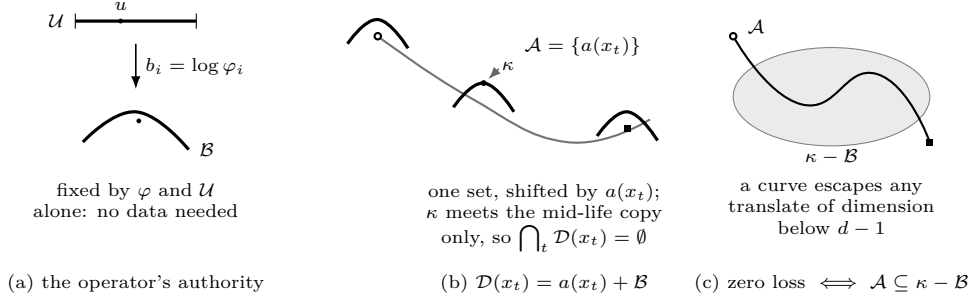


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

dynamics carry a fractional-order derivative. The fraction is not a technicality: it is the source of the trade-off, as follows.

Suppose T_f were fixed. Nothing in (2) penalises the input directly, so the answer would be trivial: find whichever admissible u minimises β and stay there. Because T_f is itself controlled, the denominator of (24) rules that solution out. The operating point that aligns the indicator is in general not the operating point that prolongs life, so aligning now shortens the interval over which the alignment is averaged. The optimum must trade the two, and Section 6.5 measures how steep that trade is on a concrete plant. A fixed-horizon formulation represents none of it, which is why one was tried and discarded (Section 6.5).

5.1. A converse bound uniform over trajectories

Before computing anything we record a certificate that removes the conditioning. Theorem 4.1 bounds the loss along a trajectory the input has already chosen; the following proposition bounds it over all of them at once, and it is what makes the number computed below a limit rather than an observation.

Proposition 5.1 (verification certificate). *Fix an indicator q and a constant $\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood of the admissible state region such that*

- (i) $V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and
- (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence over all trajectories.

Proof. Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and the chain rule

gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e., and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$ gives the claim. $\square$

Condition (25) constrains the plant data alone: it quantifies over every state and every admissible input, and mentions no trajectory. Any V one can exhibit therefore certifies a limit, and the search for a converse becomes the search for a verification function.

The C^1 hypothesis cannot be weakened to Lipschitz continuity. A Lipschitz V is differentiable almost everywhere in the state by Rademacher's theorem, but a single trajectory is a null set and could in principle run inside the exceptional set, so the chain rule step would be unavailable; recovering the argument at that generality needs the Clarke generalised gradient in place of ∇V . Smoothness is also, as it turns out, exactly what the natural candidates lack.

We tried two. The value function of the Dinkelbach problem below satisfies (25) with equality in the viscosity sense [25], but value functions of this class are only Lipschitz. Computed on a 41^3 grid and tested on a 61^3 one, (25) fails at about 45% of test points by up to 2.9×10^{-3} . Halving λ moves that violation by 0.6%, so the fault lies in the candidate and not in the bound. At the opposite extreme an affine $V(x) = c_0 - a_0^\top x$ has an exactly constant gradient and makes every requirement linear, so the best certifiable λ is a linear program; over 167,936 constraints it returns $V \equiv 0$ and only the pointwise bound $\min_{x,u} \ell$, which is vacuous here because the reachable floor falls to within sampling resolution of zero at mid-life. Deleting the two boundary requirements lets that program run to any cap imposed on it, which identifies them as what binds: trajectories of (2) increase in every coordinate, so an affine V is monotone along all of them at once and $V(x_0) \geq 0 \geq V|_{\text{failure}}$ forces it to decrease, while the near-aligned states, where the running cost vanishes, demand a local increase.

The two failures are complementary, and together they say what a certificate needs. The value function varies its gradient exactly as the two demands require but is not differentiable where that matters; affine functions are perfectly smooth but cannot vary their gradient at all. A certificate must therefore be both smooth and non-constant in its gradient, which points at a smooth under-approximation of the value function, obtained by sum-of-squares programming when the plant data are polynomial, or by interval arithmetic with an explicit Lipschitz constant. We do not claim a certified converse here, and the limit below remains a computation; what Proposition 5.1 buys is the replacement of an open problem without shape by a concrete object with a checkable property.

Problems of the form (24) are handled by Dinkelbach's transform [26]: setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman equation on the state alone, with no time argument. Because $\dot{x} > 0$ the state advances monotonically, and a semi-Lagrangian scheme [27] converges by value iteration. One implementation detail matters: the scheme must use a fixed spatial step rather than a fixed time step, because degradation speeds vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at one end of the envelope or the other.

6. Numerical study

Sections 3–5 are analytical, and three of their claims are stated sharply enough to be tested rather than illustrated: that the loss is invisible to a Fisher criterion, that the counting rule $m \geq d - 1$ binds in both directions on a generic plant, and that the best achievable loss is a finite number one can compute. This section instantiates the model on a synthetic three-mechanism plant on which every quantity is available in closed form, checks those three claims against it, measures what steering is worth, and reads off the dimensionless index that says in advance whether it will repay the effort. Section 7 then reports what measured fleets can and cannot establish about the premises.

6.1. The test plant

We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is the reciprocal temperature $1/(kT)$, so that this is the Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms. There is a single

Table 1: Loss of the fixed indicator at a Fisher score held equal. The duty cycle δ holds $\|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts zero loss out of reach unless this plant is one of the exceptions of Remark 4.5; Section 6.3 computes the floor and finds that it is not. Noise is unit and the signal-to-noise weight of Lemma 2.1 is $w = 0.9$. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of (3). The two values are equal by coincidence and play unrelated roles: w fixes the signal-to-noise ratio at which the loss is evaluated, x_f fixes where each trajectory stops. Raising x_f to unity leaves every qualitative conclusion below intact and moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° , and the gain of Section 6.5 falls from 68 to 58. No constant reported below should be read across thresholds.

Verification of the identities. Before using them we checked the analytical results numerically. The rotation identity (20) holds to relative error below 1.1×10^{-7} over the five input–channel pairs $(d, m) = (2, 1), (3, 2), (3, 1), (4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; Theorem 3.6 to 5.1×10^{-6} , limited by finite differencing; and (17) to 1.1×10^{-16} .

6.2. A Fisher level set of varying loss

Proposition 4.6 is easy to demonstrate on this plant. For a constant input, the duty cycle only reparametrises time, so it moves $\|\mathbf{d}\|$ while leaving $\hat{\mathbf{d}}(\tau)$, and hence β , untouched: the Fisher score and the loss can be varied independently. Table 1 does exactly that, holding the Fisher score equal to seven digits by tuning the duty cycle δ . The loss still spans a factor 9.4.

The two objectives also disagree on which input to prefer: the raw score, before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and the loss is 1.7 times its value at $u = 1$. How far they disagree is a property of the instance; what is structural is only the blindness.

6.3. The reachability floor with a single input

With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} = 4.52^\circ$; it falls through mid-life to below the 0.01° resolution of our sampling, where the indicator is momentarily reachable, and rises again towards failure (Fig. 5(a)). Zero average loss would need the indicator reachable at every instant and not at

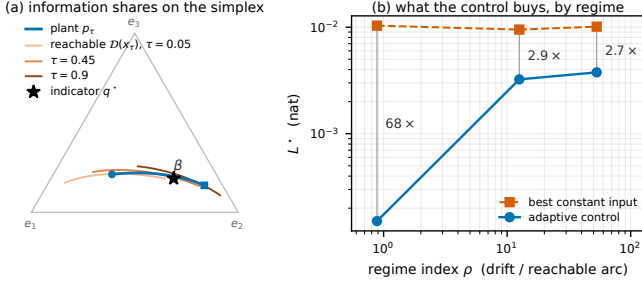


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against the regime index ρ .

one, so it is unattainable however the input is steered, in agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable set sweeps across q^* as the plant ages, so the mid-life arc passes close to the indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

6.4. The counting rule, tested in both directions

Theorem 4.1 is an equivalence, and the paragraph above tests only half of it: with too few inputs the floor stays positive. The other half, that supplying the inputs the counting rule asks for makes the floor vanish, is the more useful claim for a practitioner deciding how many operating variables to instrument, so we test it directly.

Give the same three-mechanism plant a second independent input by writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$ and each input acts on the mechanisms in a different pattern. Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns, with a coefficient matrix of determinant 0.27 here, hence a unique solution at every state.

Taking as target the direction the plant occupies at mid-life under a central operating point, that solution lies inside the admissible box $[0, 3]^2$ at every one of 600 sampled states covering the whole life, and the residual misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision; the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid search over the box returns a floor of at most 0.156° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero at every sampled state on the addition of one operating variable. On this plant the counting rule is therefore not a conservative sufficient condition but a sharp one, and within the adopted model it fixes how many independent control degrees of freedom the plant must offer. How those degrees of freedom correspond to physical actuators, and how the mechanisms are individuated in the first place, are modelling questions this paper

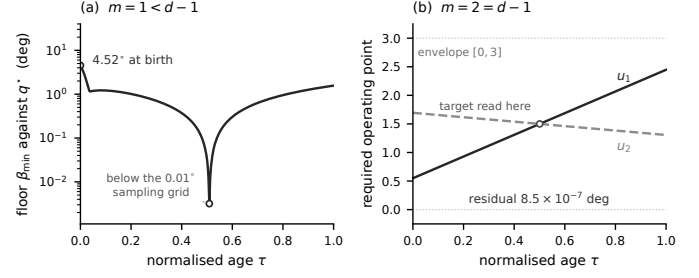


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

does not settle. Proposition 4.4 says which plants escape it, and Remark 4.5 says why a designer should not count on being one of them.

6.5. The value of steering

Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive policy attains 1.51×10^{-4} against 1.03×10^{-2} under the best constant input, a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy, rising to 72 at 161 points; we report it on the 41-point grid used throughout.

The factor is an information ratio and is not free in reliability terms: the policy that attains it runs the component differently, and Section 6.6 finds the two closed loops separated by a factor of 3.4 in life. Aligning the indicator and prolonging the component are competing objectives, which is what makes the horizon in (24) a controlled quantity rather than a constant. That factor is the value of steering at all, not the value of steering optimally: a myopic policy that attends only to the numerator of (24) sits 2.7% above the optimum at q^* .

The fraction in (24) is doing visible work at this operating point. The closed loop at q^* and the closed loop at a balanced indicator, which calls for far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the numerator is paid for in the denominator. This is the tension anticipated in Section 5, and it is why the horizon cannot be fixed in advance.

6.6. The fractional limit

Dinkelbach bisection with the control grid matched to the policy gives, over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.47 \times 10^{-4}$, below the realisable 1.51×10^{-4} as it must be.

We quote this to two digits only, for the following reason: the raw values are twice the extrapolant, so the first-order correction carries half the answer, and both quantities still move by 4% between control grids of 41 and 81 points. Three checks guard the number: F must decrease and cross zero, λ^* must not exceed a realisable policy, and the grid sequence must extrapolate consistently. A fixed-horizon formulation failed the second and third and was

Table 2: The regime index. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	L^*	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.5×10^{-4}	$68 \times$
(0.80, 0.82, 0.79)	1.93°	12.5	3.3×10^{-3}	$2.9 \times$
(0.80, 0.805, 0.798)	0.45°	53.1	3.8×10^{-3}	$2.7 \times$

discarded, as did a first attempt with mismatched control grids. A converse bound uniform over trajectories which would replace this computation by a theorem, remains open.

6.7. The regime index as a screening test

For a practitioner the first question is not how to steer but whether to bother. One dimensionless ratio answers it. The reachable set $\mathcal{D}(x)$ is an arc; write ρ for the drift of its midpoint over the life divided by its mean length, so ρ measures how far the informative direction wanders in units of the actuation authority available to chase it.

Table 2 varies the spread of the activation energies, which by (20) is what sets that authority. As the colapses, the attainable loss L^* rises by a factor 25 and the value of control falls from 68 to 2.7: with weak authority there is little left to steer. Fig. 4(b) plots the two against ρ . The construction of this paper is a statement about the low- ρ regime, and ρ can be computed from a nominal degradation model before any indicator is designed.

7. What public data can say about the premises

The construction rests on three modelling choices, and each is empirically testable. They do not carry equal weight, and Fig. 6 separates what depends on which. Two are premises about the plant: that the informative direction rotates along the trajectory, and that the input moves it. The third is the separability of Lemma 4.3, which is what makes the operator’s authority computable offline. This section summarises what six public run-to-failure fleets currently settle: the Severson lithium-ion cells [28], the XJTU-SY [29] and PRONOSTIA [30] bearing fleets, the NASA battery cells [31], and the C-MAPSS [32] and N-CMAPSS [33] turbofan simulations. The protocols are not reproduced here; the datasets are named so that each test can be located and repeated.

Rotation. On 129 Severson lithium-ion cells [28], with capacity fade, resistance growth and mean temperature as the three mechanisms, the fleet-mean informative direction travels 30.7° of arc over $\tau \in [0.15, 0.9]$ and ends 21° from where it began (Fig. 2(c)). The direction stays in the open positive orthant once the differentiating window reaches 301 cycles, so the positivity that Remark 3.1 shows to be load-bearing is a property of this fleet and not only of the model.

Actuation. This premise needs the operating point to vary while everything else is held fixed, which of the six fleets only the two turbofan simulations permit. On C-MAPSS [32], whose regimes interleave cycle by cycle inside each engine so that an engine serves as its own control, the angle between half-samples of different regimes exceeds the angle between half-samples of the same regime by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines. Against fake regimes cut from a single real one, where no contrast exists by construction, the same machinery returns $|B - W| \leq 0.5^\circ$. On the four measured fleets the obstacle is the estimator rather than the physics: in the two bearing fleets the operating condition is fixed per unit, so “condition” and “group of units” cannot be separated, and the NASA fleet has four cells. The premise is therefore confirmed on simulation and open on hardware.

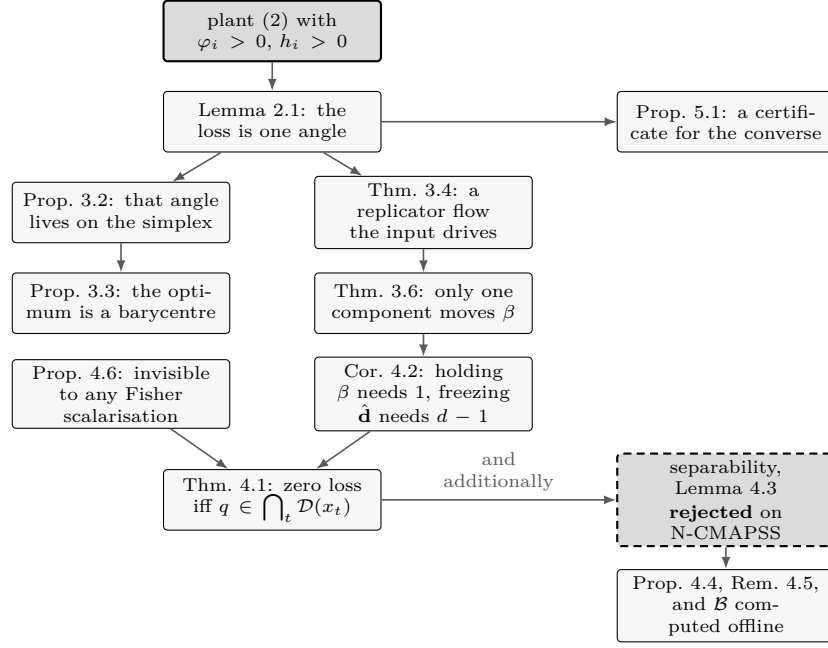
Separability. Lemma 4.3 is a statement about a plant class, and data can refute it. If the shift between two operating regimes were the same at every stage of life, regime and stage would act additively on ξ and a two-way layout would show no interaction. On N-CMAPSS [33], which records altitude, Mach number and throttle every second of real flight profiles, the interaction exceeds both main effects on the sums of squares, by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, and remains larger under every clustering and staging tried. Separability is therefore rejected on the highest-fidelity public turbofan simulation.

The consequence is the scope of the results above. The geometry of Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What separability buys, and what the rejection withdraws, is Proposition 4.4 and the offline route to $\mathcal{B}$: those describe a model class, and a plant must be shown to belong to it before they are used. The two-way layout above is the test, and it costs one pass over data a reliability programme already holds.

8. Discussion

Four practical readings follow from the results above. The first two hold whatever the plant is, because one is a test and the other an invariance. The last two assume the separable form of (2), which Section 7 reports as rejected on the only data able to test it, so they apply where that form does and not otherwise. The distinction matters more than any of the four.

Test the separability assumption before relying on it. The two-way layout of Section 7 needs only a fleet whose operating point varies within a unit, and it returns an interaction against a measured noise floor. If the interaction clears that floor, the stress model is not separable on that fleet, the offline route to the operator’s authority is unavailable, and the two model-based readings below do not apply. The test costs one pass over data the programme already holds.



Every result on the spine needs only the positivity of the degradation model. Only the dashed branch needs separability, and it is that hypothesis, not the geometry, that Section 7 reports as rejected on the best public turbofan data.

Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

A Fisher criterion cannot be used to design a fixed indicator. This is the second of the two unconditional readings: Proposition 4.6 is an invariance of the loss under common gain, and holds whatever the plant. Design rules that maximise a scalarisation of the Fisher information, natural and common choice, are provably unable to distinguish configurations whose losses differ by an order of magnitude, because they measure the strength of the signal and not the direction it points in. If the two objectives coincide in a given application, that is a property of the application and should be verified, not assumed.

The binding constraint is actuation authority, not indicator sophistication. The counting rule of Theorem 4.1 prices the operating variables a lossless fixed indicator would need, and Proposition 4.4 sharpens it: the count is sufficient and not necessary, but the exceptions ask the plant's log-offsets to slide along the input's own direction, which Remark 4.5 shows to be a knife edge. Care in choosing the weights substitutes for actuation in neither case. What one controllable variable does buy is the weaker objective of Corollary 4.2: an installation that could never freeze the informative direction can still hold the misalignment constant, though not at zero. Authority bought this way is not free either. On the test plant the indicator whose loss is smallest is served by a policy that changes the component's life by a factor of 3.4 against a balanced indicator, so information-preserving operation and long life are competing objectives and the trade between them is a reliability decision rather than an estimation one.

The decision to steer precedes the design of the indicator. The regime index ρ of Section 6.7 requires only a nominal degradation model and the operating envelope. When ρ is large, the reachable arc is short relative to the drift, and Table 2 shows the value of steering collapsing from $68\times$ to under $3\times$. The index is therefore a screening test to run before committing engineering effort.

Limitations. Three limitations qualify the readings above. The separable form of (2) is not only an idealisation but one the data reject: Section 7 finds the interaction between operating point and state larger than either main effect on two N-CMAPSS datasets, so computing the operator's authority offline is available only where that form holds. The fundamental limit of Section 5 is computed rather than proved, to two digits. And the actuation premise is confirmed on simulated rather than measured data, for the reasons set out in Section 7.

9. Conclusion

A fixed health indicator discards information, and this paper has treated what it discards as a quantity the operating point controls. Indicator design then becomes a reachability question. The family of reachable direction sets bounds what the operating point can achieve, and their drift relative to their width is a dimensionless measure of whether substantial benefit is plausible. That ratio is dimensionless and available before the designer fixes any

indicator. Actuation authority, rather than sophistication of the indicator, governs whether a fixed compression can come close to lossless. Whether that authority is present is an empirical question rather than a modelling assumption, and on 439 simulated turbofan records it is. Establishing the same on measured fleets awaits records in which the operating point varies enough to show the effect.

The separable plant form that makes all of this computable is rejected by the data, on two datasets of the highest-fidelity public turbofan simulation, so the geometry of Section 4.1 describes a model class rather than measured machinery, and the offline route it opens is available only where that class applies. What replaces it when the stress model is not separable, and whether a weaker structure still yields a computable reachability criterion, is the question this leaves.

Two further questions remain open, the first of them now in a more specific form. Proposition 5.1 reduces the search for a converse bound uniform over trajectories to the exhibition of a single continuously differentiable verification function; the value function is the natural candidate and fails the differential test for a structural reason, so what is required is a smooth under-approximation of it, by sum-of-squares programming or by interval arithmetic with an explicit Lipschitz constant. The other is unchanged: the fractional limit to better than two digits, which requires a scheme of higher order than the semi-Lagrangian one used here.

Data availability and reproducibility

The six fleets named in Section 7 are publicly available. One script regenerates every computed result from a single configuration, and a companion check recomputes each reported number and compares it with the value written in the text, covering the tables and the quantities reported alongside them. Those checks are maintained against the full manuscript from which this paper is drawn, and all of them pass; the numbers reported here are carried over from it unchanged. Three groups of numbers lie outside it: the residuals of the identity checks in Section 6, the barycentre comparison in Section 3.3, and the diagnostics of the two discarded formulations in Section 5. The plots are produced by three further scripts. All of these are available from the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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